

Hamiltonian Edge Separation in Barnette Graphs: Certified Extremal Values and an Exact Double-Ladder Formula

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Abstract

We study the Hamiltonian strong edge-separation number $\text{hsep}(G)$. Abstractly, it is the minimum size of a completely separating subfamily of the Hamiltonian-cycle hypergraph of G : for every ordered pair of distinct edges (e, f) , some selected cycle contains e and avoids f . An exhaustive certificate audit covers all 22,263 non-isomorphic Barnette graphs through order 36. Exact upper and lower certificates establish every maximizer, while verified strict upper certificates exclude all competitors. At orders divisible by four in this range, the unique maximizers form a double-ladder family related by a four-vertex square insertion. We define this family, classify its Hamiltonian cycles, and prove for all odd $a, b \geq 3$ that

$$\text{hsep}(D(a, b)) = \frac{(a+2)(b+2) - 1}{2}.$$

Matching symbolic constructions prove the two bounds: a packing of ordered edge requirements gives the lower bound, while a boundary-plus-parity family of Hamiltonian cycles gives the upper bound through an antichain of incidence signatures. Independently verified, prediction-locked computations confirm three larger family members. These checks do not establish order-level extremality beyond the census range; whether balanced double ladders remain globally extremal after order 36 is open.

1 Introduction

Separating systems encode elements by incidences with selected tests. Ordinary separation requires distinct codes, while complete separation requires incomparable codes. In this paper the ground set is the edge set of a graph and every admissible test is a Hamiltonian cycle. This restriction couples a classical set-system condition to global spanning structure.

We investigate the resulting parameter on Barnette graphs: simple, planar, bipartite, cubic, 3-connected graphs. Four levels of evidence are kept distinct:

- (i) ordinary mathematical propositions proved in the text;
- (ii) finite computational results supported by explicit, independently checkable certificates;
- (iii) prospective computations whose values were frozen before evaluation; and
- (iv) open extremal conjectures beyond the certified census range.

The computation audits all 22,263 non-isomorphic Barnette graphs through order 36 and determines every extremal value in the nonempty censuses. Exact upper certificates are separating families; lower certificates are packings or exhaustive exclusions over certified complete Hamiltonian-cycle universes. No greedy cover size is reported as exact. Table 1 records the sequence. At orders divisible by four, the unique maximizers form the double-ladder family shown in Figure 2; the tied orders congruent to two modulo four have a different face structure.

The finite pattern leads to an infinite-family theorem. A rail/rung recurrence classifies all Hamiltonian cycles of $D(a, b)$ for odd $a, b \geq 3$. A symbolic packing supplies a lower bound for hsep , while a boundary-plus-parity family of Hamiltonian cycles supplies a matching upper bound through an incidence-signature antichain. Consequently,

$$\text{hsep}(D(a, b)) = \frac{(a+2)(b+2) - 1}{2} \quad (a, b \geq 3 \text{ odd}).$$

This determines the invariant on the infinite family, but not its extremality among all Barnette graphs of the same order. Prediction-locked checks confirm three larger instances; global extremality of the balanced family beyond order 36 remains open.

2 Definitions and certificate principles

All graphs in this manuscript are finite and undirected. A Hamiltonian cycle is regarded as an undirected edge set; reversal and cyclic rotation do not produce new cycles.

Definition 2.1 (Barnette graph). A *Barnette graph* is a simple, planar, bipartite, cubic, and 3-vertex-connected graph.

For every graph G , let $\mathcal{H}(G)$ denote its set of undirected Hamiltonian cycles; this set is empty when G is non-Hamiltonian.

Definition 2.2 (Hamiltonian strong edge separation). For distinct edges $e, f \in E(G)$, a Hamiltonian cycle C *covers* the ordered requirement (e, f) if $e \in C$ and $f \notin C$. A family $\mathcal{C} \subseteq \mathcal{H}(G)$ is *Hamiltonian edge-separating* if it covers every ordered pair of distinct edges. Define

$$\text{hsep}(G) = \min \left\{ |\mathcal{C}| : \begin{array}{l} \mathcal{C} \subseteq \mathcal{H}(G), \\ \text{for all distinct } e, f \in E(G), \text{ some } C \in \mathcal{C} \text{ has } e \in C, f \notin C \end{array} \right\}. \quad (1)$$

If no Hamiltonian edge-separating family exists, the minimum in equation (1) is defined to be ∞ . We call $\text{hsep}(G)$ the *Hamiltonian strong edge-separation number*; this terminology and the notation hsep are manuscript-specific. Abstractly, it is the minimum cardinality of a completely separating subfamily of the Hamiltonian-cycle hypergraph of G .

Definition 2.3 (Barnette extremal sequence). For every order at which at least one Barnette graph exists, put

$$M_B(n) = \max\{\text{hsep}(G) : G \text{ is a Barnette graph and } |V(G)| = n\}. \quad (2)$$

The maximum is taken in the extended nonnegative integers, so it remains well defined even if some graph has $\text{hsep}(G) = \infty$. If no Barnette graph of order n exists, $M_B(n)$ is undefined. In the certified range every graph in every nonempty census is Hamiltonian, every graph admits a separating family, and all displayed values are finite.

Given an ordered family $C_1, \dots, C_k$, associate with an edge e its incidence signature

$$\sigma(e) = \{i : e \in C_i\} \subseteq \{1, \dots, k\}.$$

The following antichain equivalence for completely separating systems is classical; see Dickson (1969); Spencer (1970) for the set-system context. We include its specialization and direct proof for Hamiltonian-cycle tests.

Proposition 2.4 (Signature antichain). *A family $C_1, \dots, C_k$ is Hamiltonian edge-separating if and only if the edge signatures form an antichain: for all distinct e, f , neither $\sigma(e) \subseteq \sigma(f)$ nor $\sigma(f) \subseteq \sigma(e)$.*

Proof. The requirement (e, f) is covered exactly when some index belongs to $\sigma(e) \setminus \sigma(f)$. Covering both ordered requirements (e, f) and (f, e) is therefore equivalent to incomparability of the two signatures. $\square$

In particular, if G has m edges and k selected cycles, the standard Sperner antichain bound gives the necessary inequality

$$m \leq \binom{k}{\lfloor k/2 \rfloor}.$$

This classical counting obstruction, also underlying separating-system bounds (Katona, 1966; Dickson, 1969; Spencer, 1970), is kept separate from graph-specific optimality certificates.

Observation 2.5 (Perfect-matching formulation). In a cubic graph, the complement of a Hamiltonian cycle is a perfect matching. Consequently, Hamiltonian strong edge separation can equivalently be encoded by perfect matchings whose complementary 2-factors are Hamiltonian cycles, with every edge-incidence bit reversed. Bitwise reversal preserves pairwise incomparability (while reversing containment), so the antichain condition is unchanged. No claim is made that this restricted matching formulation has previously been studied.

Lemma 2.6 (Primal and packing certificates). *If k explicit Hamiltonian cycles cover every ordered edge-pair requirement, then $\text{hsep}(G) \leq k$. If there are k requirements such that every Hamiltonian cycle covers at most one of them, then $\text{hsep}(G) \geq k$. Consequently, matching certificates prove $\text{hsep}(G) = k$.*

Proof. The first statement is the definition. For the second, each of the k requirements must be assigned to a selected cycle that covers it, while no selected cycle can receive two requirements. Hence at least k cycles are necessary. $\square$

When no matching packing is available, a lower bound in this project is called independently certified only if a standalone verifier checks a complete Hamiltonian-cycle universe and exhaustively excludes every cover below the claimed value.

3 Related work

Separating systems and antichains. Separating systems on a finite ground set are classical (Katona, 1966; Dickson, 1969). In an ordinary separating system, the incidence signatures of distinct elements need only be distinct. Complete separation requires, for every ordered pair (e, f) , a test containing e and avoiding f . Equivalently, all incidence signatures must be pairwise incomparable, connecting the problem directly with Sperner theory and minimal completely separating systems (Dickson, 1969; Spencer, 1970). The signature argument in Proposition 2.4 is precisely this classical equivalence after the admissible tests have been restricted to Hamiltonian cycles.

Graph-structured separators. Graph-specific separating systems have a longer history; Cai (1984), for example, studied separating systems of graphs. We cite that work only as related graph-specific context and do not assert that its definition coincides with hsep . A substantial line of work asks for edge separation by paths (Falgas-Ravry et al., 2014; Balogh et al., 2016; Bonamy

et al., 2023; Letzter, 2024). These problems use the same or a closely related directed separation condition, but admit paths rather than Hamiltonian cycles as tests. Recent cycle-separation work allows arbitrary cycles or subdivisions of K_4 (Botler and Naia, 2025); this is a much larger admissible family than the Hamiltonian cycles of a fixed graph. Cycle-identification problems, such as those studied for hypercubes and tori by Honkala et al. (2003), generally seek distinct nonempty signatures. Distinctness is weaker than the pairwise incomparability required by hsep.

Hamiltonicity and Barnette graphs. Hamiltonian cycles in cubic 3-connected bipartite planar graphs were studied computationally and structurally by Holton et al. (1985). Because the complement of a Hamiltonian cycle in a cubic graph is a perfect matching, Observation 2.5 also places the present parameter near matching formulations of Barnette’s conjecture (Gorsky et al., 2023). Broader perspectives on that conjecture are given by Alt et al. (2016). Our finite census uses the planar-graph generation methods of Brinkmann and McKay (2007) and Plantri 5.8 (Brinkmann et al., n.d.).

The underlying notions of complete separation, antichain signatures, and edge separation by unrestricted paths or cycles are therefore not new. The specific restriction here is that every admissible test must be a Hamiltonian cycle of the given graph. To the best of our knowledge, strong edge separation under this restriction has not previously been investigated; this cautious assessment is not an absolute novelty claim.

4 Computational method

4.1 Census generation and graph validation

Complete non-isomorphic censuses were generated with plantri 5.8 (Brinkmann et al., n.d.). For target order n , the dual generation command has the form

$$\text{plantri -b -c3 -d T -}, \quad T = n/2 + 2.$$

The binary `planar_code` stream retains the rotation system. Each record was checked for the requested order and edge count, simplicity, cubicity, bipartiteness, planarity, 3-connectivity, and Euler and face arithmetic. Canonical hashes were computed from the sphere embedding; graph6, planar code, canonical edge lists, rotations, and faces were retained in the certificates. Generation indices were required to be contiguous and canonical hashes unique. Census completeness was never inferred from a retained candidate subset.

The plantri command, version, executable digest, environment, worker count, and runtime were recorded per order. At most four workers were used. Expensive stages used deterministic ordering, atomic checkpoints, and resumable output.

4.2 Hamiltonian cycles and separating covers

For a cubic graph of order n , there are $m = 3n/2$ edges and therefore $m(m - 1)$ ordered edge-pair requirements. At order 36 this is $54 \cdot 53 = 2,862$ requirements per graph. Every emitted cycle was checked for closure, vertex uniqueness, spanning, and edge validity before it was used as a set-cover column.

For the smaller censuses, the complete undirected Hamiltonian-cycle universe was enumerated for every graph and the set-cover problem was solved exactly. For larger censuses, the pipeline first constructed deterministic separating covers for every graph and ranked the difficult tail. Once a candidate value k had an exact lower and upper certificate, a competitor needed only a verified cover

of size at most k , or at most $k - 1$ to establish uniqueness. Complete-universe enumeration and exact optimization were then restricted to graphs that could still attain the maximum. A timeout or incomplete universe was not accepted as a conclusion.

4.3 Three levels of evidence

The proof pipeline deliberately separates the following objects.

Solver output. A proposed cycle, cover, optimum, or infeasibility result produced by a search or optimization implementation.

Emitted certificate. Canonically encoded graphs and explicit cycles, together with either packing requirements or a certified complete cycle universe suitable for exhaustive exclusion.

Independent verification. A standalone Python standard-library program reparses the artifacts, checks graph and cycle structure, evaluates every ordered requirement, checks lower certificates, order summaries, and SHA-256 manifests, and does not import the project, NetworkX, plantri, SAT, or optimization packages.

The standalone verifier checks the emitted census records and their internal consistency; it does not independently prove the enumerative completeness of plantri. The finite census-completeness claim rests on the recorded complete plantri invocation together with contiguous output and the subsequent record audits.

For every nonmaximizer the final package contains an independently verified cover strictly smaller than the maximum at that order. For every maximizer it contains an exact primal certificate and either a matching packing or a complete-universe exhaustive lower proof. Thus a greedy witness count is never reported as $\text{hsep}(G)$.

5 The certified finite extremal sequence

Computational Result 5.1 (Certified values through order 36). The complete Plantri 5.8 censuses and independent certificate verification give the values in table 1. Across the nonempty orders, 22,263 non-isomorphic graphs were audited. Every maximum and tie count was independently verified, and every nonmaximizer has a verified separating cover strictly below the maximum of its order.

Every cubic graph has even order, since the degree-sum identity gives $3|V| = 2|E|$. Hence all odd orders from 9 through 35 are impossible. The complete order-10 Plantri census is empty. These undefined entries are not optimization failures.

The unique order-36 maximizer has $\text{hsep} = 49$ and 68 Hamiltonian cycles. A 49-cycle primal certificate and a 49-requirement packing certificate prove its exact value, while all 15,373 competitors have independently checked covers of size at most 48. Its canonical identifier is recorded in the reproducibility supplement.

Matching packings certify the displayed maximizers at orders 8, 12, 16, 20, 24, 28, 32, 36. At orders 14, 18, 22, 26, 30, 34, at least one displayed exact result uses exhaustive set-cover exclusion over a certified complete cycle universe instead. This distinction concerns the form of the lower certificate, not the exactness of the result.

Status note. Computational Result 5.1 is an exhaustive finite computational theorem about the listed orders. It is not a proof of any formula for arbitrary n , nor does it assert anything about orders 38 or 40.

Table 1: Certified Barnette Hamiltonian edge-separation extrema. The last column is the number of undirected Hamiltonian cycles in each maximizer; within every tied order all maximizers have the displayed count.

n	graphs	$M_B(n)$	maximizers	Hamiltonian cycles
8	1	6	1	6
12	1	8	1	8
14	1	10	1	12
16	2	12	1	14
18	2	10	1	16
20	8	17	1	20
22	8	13	2	28
24	32	24	1	30
26	57	18	5	40
28	185	31	1	40
30	466	25	3	60
32	1,543	40	1	54
34	4,583	32	7	80
36	15,374	49	1	68

6 The double-ladder family

Let $L_a = P_{a+1} \square K_2$ be a ladder with vertices A_i^r , where $0 \leq i \leq a$ and $r \in \{0, 1\}$. Its rungs are $A_i^0 A_i^1$, and its rails are $A_i^r A_{i+1}^r$. Define a second, disjoint ladder L_b on vertices B_j^r .

Definition 6.1 (Double ladder). For positive integers a, b , the graph $D(a, b)$ consists of L_a and L_b together with the four connector edges

$$A_0^0 B_0^0, \quad A_0^1 B_b^0, \quad A_a^0 B_0^1, \quad A_a^1 B_b^1. \quad (3)$$

Thus

$$|V(D(a, b))| = 2(a + b) + 4.$$

6.1 The square insertion

Suppose $uvyxu$ is a facial quadrilateral, so that uv and xy are opposite edges. Replace uv by $upqv$, replace xy by $xrsy$, and add pr and qs . The four new vertices induce the facial cycle $pqsrp$. In a ladder cell this operation replaces one cell by three and increases the ladder length by two.

Lemma 6.2 (Preservation under facial square insertion). *Let G be a simple, planar, bipartite, cubic, 3-connected graph, and apply the preceding operation to opposite edges of a facial quadrilateral. The resulting graph G^+ has the same five properties.*

Proof. The four terminals are distinct and the inserted vertices are new, so simplicity is immediate. Drawing the three-cell ladder inside the old face proves planarity. Each old terminal loses one edge and gains one edge; every new vertex has two neighbours on the inserted 4-cycle and one old terminal. Thus cubicity is preserved. If the facial quadrilateral is $uvyxu$, then u, y lie in one bipartition class and v, x in the other. The unique alternating coloring along the two new length-three paths makes both pr and qs bichromatic, proving bipartiteness.

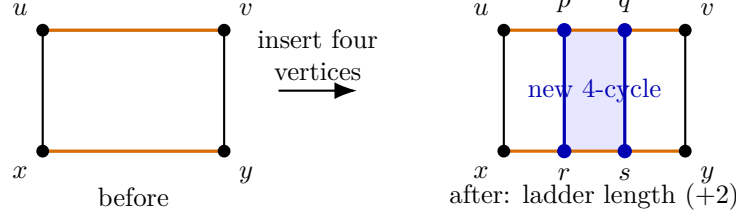


Figure 1: Local square insertion. The opposite facial edges (uv) and (xy) (orange, left) are replaced by the paths $(upqv)$ and $(xrsy)$. The new edges (pr) and (qs) complete the quadrilateral $(pqsrp)$.

It remains to prove 3-connectivity. Let $S \subseteq V(G^+)$ with $|S| \leq 2$, and call p, q, r, s the new vertices. If S contains no new vertex, paths in the connected graph $G - S$ lift through the two intact replacement paths; the intact inner 4-cycle is attached to every surviving terminal. If S contains exactly one new vertex, at most one old vertex is deleted. The graph obtained from G by deleting that old vertex is 2-connected and hence has no bridge. Delete, if necessary, the one old edge whose replacement path contains the removed new vertex. The remaining old vertices are still connected, and the other three new vertices form a path attached to at least two surviving terminals. Finally, if two new vertices are deleted, no old vertex is deleted. A 3-vertex-connected graph is 3-edge-connected, so $G - \{uv, xy\}$ is connected. Every remaining new vertex has its own terminal edge into this connected old core. Hence $G^+ - S$ is connected in all cases, and G^+ is 3-connected. $\square$

Theorem 6.3. *For all odd integers $a, b \geq 1$, the graph $D(a, b)$ is a Barnette graph.*

Proof. The graph $D(1, 1)$ is isomorphic to the cube. Increasing either odd parameter by two is exactly the facial square insertion of Lemma 6.2 in a cell of the corresponding ladder. Starting from $D(1, 1)$, repeated insertions therefore reach every pair of positive odd parameters while preserving all Barnette properties. $\square$

6.2 Certified occurrences and face structure

Computational Result 6.4 (Certified double-ladder recurrence). Exact graph isomorphisms identify the unique maximizers at orders divisible by four as follows:

n	8	12	16	20	24	28	32	36
G_n	$D(1, 1)$	$D(3, 1)$	$D(3, 3)$	$D(5, 3)$	$D(5, 5)$	$D(7, 5)$	$D(7, 7)$	$D(9, 7)$.

All seven consecutive transitions have reconstructible expansion certificates whose reconstructed edge set is exactly the certified target and whose reduced canonical hash is exactly the certified source.

Proposition 6.5 (Face pattern in the recursive embedding). *For odd positive a, b , the embedding obtained from the cube by the square insertions above has $a + b$ designated ladder-cell faces, all quadrilateral, together with two faces of length $a + 3$ and two faces of length $b + 3$. When $a = 1$ or $b = 1$, the corresponding two cap faces are themselves quadrilateral.*

Proof. For $D(1, 1)$, the two designated cells and four cap faces are the six faces of the cube. Increasing a by two replaces one designated cell by three, adding two quadrilateral cells. The two cap boundaries running along the modified (a)-ladder each acquire two edges; the other two cap

Double-ladder family (orders above 36: global extremality unproved)

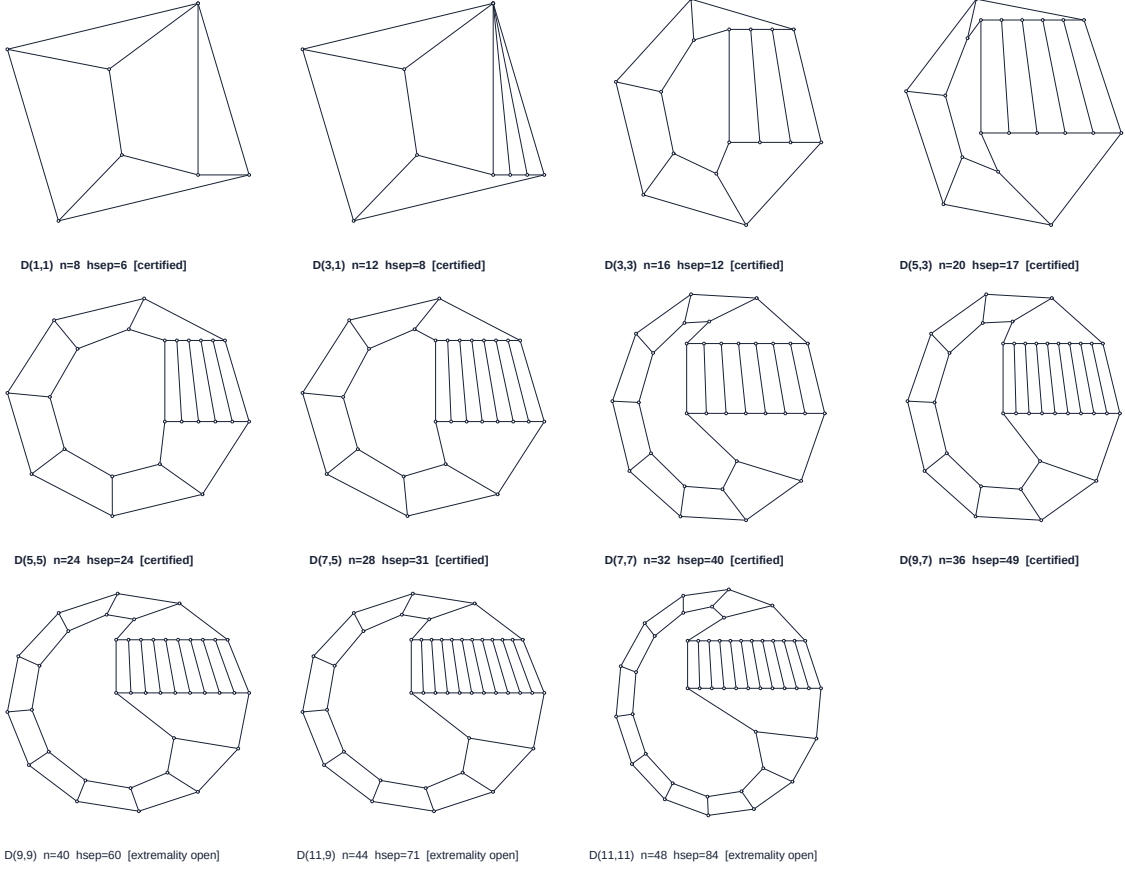


Figure 2: The double-ladder family through $D(11, 11)$. The members from $D(1, 1)$ through $D(9, 7)$ are the certified unique hsep-maximizers at the corresponding orders divisible by four through order 36. The graphs $D(9, 9)$, $D(11, 9)$, and $D(11, 11)$ are proved family members with exact graph-specific hsep values; their global extremality at orders 40, 44, and 48 is unproved.

boundaries are unchanged. This is precisely the claimed update. The argument for b is identical, so induction proves the formula. $\square$

For the certified members from $D(3, 3)$ onward, the two sets of designated cells are exactly the two path components of the quadrilateral face-adjacency graph. At orders 8 and 12, additional quadrilateral cap faces make this face-adjacency description degenerate, although the abstract double-ladder definition remains valid.

7 Hamiltonian-cycle structure

The general classification below agrees with the complete certified Hamiltonian-cycle universes of every analyzed double-ladder member, including the prediction-locked graphs beyond order 36. Those finite universes provide an independent check of the cycle counts and classes proved here.

Theorem 7.1 (Hamiltonian cycles of the double ladder). *For odd $a, b \geq 3$, the Hamiltonian cycles of $D(a, b)$ are one all-rails cycle, the ab cycles with one turn in each ladder, and four two-connector exceptions. In particular,*

$$|\mathcal{H}(D(a, b))| = ab + 5.$$

Proof. For a ladder of length m , put binary variables $x_{t,s}$ on rail s of cell t , y_t on rung t , and $c_{0,s}, c_{m,s}$ on its endpoint connectors. At an internal vertex the degree-two equation is

$$x_{t-1,s} + x_{t,s} + y_t = 2,$$

with the analogous endpoint equations including $c_{0,s}$ or $c_{m,s}$. Subtracting the equations on the two rails gives $d_t = -d_{t-1}$, where $d_t = x_{t,0} - x_{t,1}$. Thus the endpoint connector bits determine the alternating rail difference, after which every rung is forced.

If all four connectors are selected, $d_0 = 0$, so the two rail choices agree in every cell. A zero rail pair forces the two bounding rungs. Two such zero pairs enclose a separate cycle, so a connected spanning solution has either no zero pair or one. The two ladders must have the same type: one straight and one turned produces two components. Hence there is one all-rails cycle and one cycle for each independent pair of turn positions.

If exactly two connectors are selected, applying $d_t = (-1)^t d_0$ at the far endpoint and using odd m leaves the four connector words 1100, 0011, 1010, 0101. In each case the recurrence has one completion: one ladder uses all rungs and alternating rails, while the other uses all rails and only the appropriate endpoint rung. The remaining words close a component before all vertices are reached. The connector cut must be crossed a positive even number of times, so zero, one, and three connectors cannot yield a Hamiltonian cycle. These cases exhaust the recurrence. $\square$

8 Exact edge separation for double ladders

Throughout this section, let $a, b \geq 3$ be odd. In ladder A , write $r_A(p) = A_p^0 A_p^1$ for rung p , $0 \leq p \leq a$, and $\ell_A(i, s) = A_i^s A_{i+1}^s$ for rail s of cell i , $0 \leq i < a$. Use $r_B(q)$ and $\ell_B(j, s)$ analogously. Denote the all-rails cycle by R , the cycle turning in cell i of A and cell j of B by $T(i, j)$, and the four exceptional cycles from Theorem 7.1 by

$$X_{0011}, \quad X_{1100}, \quad X_{0101}, \quad X_{1010}.$$

The subscripts record connector words in the order of equation (3).

8.1 A symbolic packing lower bound

We first construct ordered edge requirements whose cycle supports are pairwise disjoint. Four requirements isolate the four exceptional cycles. One concrete choice, anchored on ladder A , is

$$(r_A(0), r_A(1)), \quad (r_A(a), r_A(a-1)), \quad (\ell_A(0, 0), \ell_A(0, 1)), \quad (\ell_A(0, 1), \ell_A(0, 0)). \quad (4)$$

The incidence recurrence in Theorem 7.1 shows directly that the four supports are distinct singleton sets, one for each X -cycle.

Next isolate every boundary turn. For $0 \leq j < b$, use

$$(r_A(0), \ell_B(j, j \bmod 2)), \quad (r_A(a), \ell_B(j, 1 - j \bmod 2)). \quad (5)$$

They have singleton supports $\{T(0, j)\}$ and $\{T(a-1, j)\}$, respectively. For $1 \leq i \leq a-2$, the symmetric requirements

$$(r_B(0), \ell_A(i, i \bmod 2)), \quad (r_B(b), \ell_A(i, 1 - i \bmod 2)) \quad (6)$$

have supports $\{T(i, 0)\}$ and $\{T(i, b-1)\}$. The parity choice in equations (5) and (6) ensures that an exceptional cycle containing the endpoint rung also contains the excluded rail, so no exceptional cycle enters a boundary support.

It remains to handle the interior turn grid. For $0 < p < a$, Theorem 7.1 gives

$$\text{supp}(r_A(p), \ell_B(j, s)) = \{T(p-1, j), T(p, j)\}, \quad (7)$$

and, symmetrically, for $0 < q < b$,

$$\text{supp}(r_B(q), \ell_A(i, s)) = \{T(i, q-1), T(i, q)\}. \quad (8)$$

Here either rail side $s \in \{0, 1\}$ may be used. These requirements are dominoes in the $(a-2) \times (b-2)$ grid of interior turns.

Delete its far corner $T(a-2, b-2)$, which belongs to the majority parity class. In each interior B -column, pair the A -rows 1–2, 3–4, $\dots$, $(a-4)$ – $(a-3)$. In the remaining row $a-2$, pair the B -columns in the same way. This tiles every remaining interior cell and leaves precisely the deleted corner.

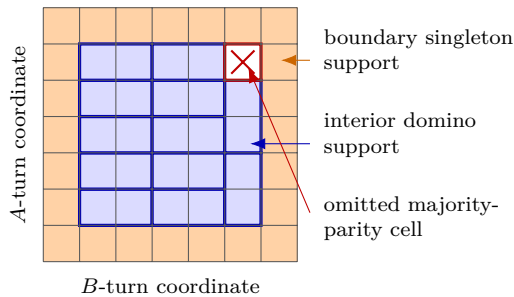


Figure 3: The packing construction, shown for a 7×7 turn grid. Orange boundary cells and the four exceptional cycles have singleton support; blue rectangles are pairwise disjoint two-turn supports. The construction for general odd a, b is the same row-first domino tiling with one majority-parity interior cell omitted.

Theorem 8.1 (Symbolic packing bound). *For odd $a, b \geq 3$,*

$$\text{hsep}(D(a, b)) \geq \frac{(a+2)(b+2) - 1}{2}.$$

Proof. The four supports in equation (4), the boundary singleton supports, and the domino supports in equations (7) and (8) are mutually disjoint: they partition all Hamiltonian-cycle classes except R and the deleted interior turn. The all-rails cycle covers none of the chosen requirements. Thus every Hamiltonian cycle covers at most one chosen requirement. Their number is

$$4 + (2a + 2b - 4) + \frac{(a-2)(b-2) - 1}{2} = \frac{(a+2)(b+2) - 1}{2}.$$

The packing half of Lemma 2.6 proves the claim. $\square$

8.2 A symbolic separating-family upper bound

Define

$$\begin{aligned} \mathcal{C}(a, b) = & \{X_{0011}, X_{1100}, X_{0101}, X_{1010}\} \\ & \cup \{T(i, j) : i \in \{0, a-1\} \text{ or } j \in \{0, b-1\}\} \\ & \cup \{T(i, j) : 0 < i < a-1, 0 < j < b-1, i+j \text{ odd}\}. \end{aligned} \quad (9)$$

There are $2a + 2b - 4$ boundary turns. The odd parity class of the odd by odd interior grid has $((a-2)(b-2) - 1)/2$ cells, and hence

$$|\mathcal{C}(a, b)| = 4 + (2a + 2b - 4) + \frac{(a-2)(b-2) - 1}{2} = \frac{(a+2)(b+2) - 1}{2}. \quad (10)$$

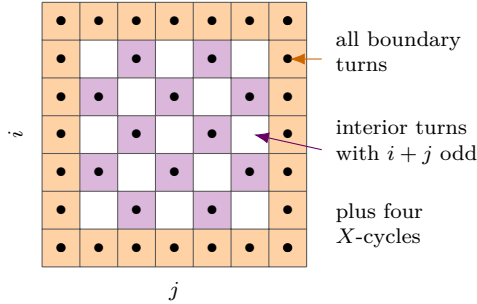


Figure 4: Turn cycles selected by $\mathcal{C}(a, b)$, again illustrated on a 7×7 grid. Every boundary turn is selected, as is one checkerboard class of interior turns; the four connector-exception cycles are added separately.

We prove that this family separates by giving both directed witnesses for every pair of edge types. Let P be its selected turn cycles. Write $R_i = \{T(i, j) \in P\}$, $K_j = \{T(i, j) \in P\}$, and

$$N_A(p) = \{p-1, p\} \cap \{0, \dots, a-1\}, \quad N_B(q) = \{q-1, q\} \cap \{0, \dots, b-1\}.$$

On the turn coordinates of an edge signature, every connector has all of P ; an A -rail in cell i has $P \setminus R_i$; an A -rung p has precisely the selected turns whose first coordinate is in $N_A(p)$. The corresponding statements for ladder B use columns.

For the four exception coordinates, abbreviate $B_0 = X_{0011}$, $B_1 = X_{1100}$, $A_0 = X_{0101}$, and $A_1 = X_{1010}$. An A -rail contains B_0, B_1 and exactly one of A_0, A_1 , determined by its side and cell parity. An internal A -rung contains A_0, A_1 ; an endpoint rung also contains one of B_0, B_1 . Interchanging A and B gives the other ladder. In connector order equation (3), the four exception parts of the connector signatures are

$$\{B_1, A_1\}, \quad \{B_1, A_0\}, \quad \{B_0, A_1\}, \quad \{B_0, A_0\}. \quad (11)$$

Theorem 8.2 (Symbolic separating family). *For odd $a, b \geq 3$, the edge signatures induced by $\mathcal{C}(a, b)$ are pairwise incomparable. Consequently,*

$$\text{hsep}(D(a, b)) \leq \frac{(a+2)(b+2)-1}{2}.$$

Proof. We exhaust the eight edge-type combinations. In every case we identify a cycle in each of $\sigma(e) \setminus \sigma(f)$ and $\sigma(f) \setminus \sigma(e)$.

1. *Two connectors.* The four sets in equation (11) are distinct and have size two. Each of two such sets therefore has an exception coordinate absent from the other.
2. *A connector and a rail.* A selected boundary turn in the rail's omitted row or column belongs to the connector but not the rail. A rail has three exception coordinates—both opposite-ladder exceptions and one alternating same-ladder exception—whereas a connector has only two; the unused opposite-ladder exception belongs to the rail alone.
3. *A connector and a rung.* Since $a, b \geq 3$, there is a boundary row or column outside the rung's one- or two-cell neighbourhood. A boundary turn there belongs only to the connector. The rung contains both same-ladder exceptions, while the connector contains exactly one of them, so the other exception gives the reverse witness.
4. *Two rails.* The two sides of one cell have the same turn coordinates but opposite alternating exception coordinates. Rails in different cells of one ladder are distinguished in both directions by selected boundary turns in their two omitted rows or columns. For rails on different ladders, choose a boundary point in the omitted column but outside the omitted row, and then one in the omitted row but outside the omitted column. Both are selected and give the two directions.
5. *Two rungs of one ladder.* If their neighbourhoods are not nested, a selected boundary turn in each set difference gives both witnesses. The only nested pairs are $N_A(0) \subset N_A(1)$ and $N_A(a) \subset N_A(a-1)$, and their B -analogues. There the endpoint rung has an additional exception coordinate, while a turn in the internal rung's extra neighbouring cell gives the reverse witness.
6. *Rungs of different ladders.* Their two-element exception cores are $\{A_0, A_1\}$ and $\{B_0, B_1\}$. An endpoint rung can add at most one coordinate from the other core, so each signature retains an exception absent from the other.
7. *A rail and a rung of the same ladder.* The rail contains both opposite-ladder exceptions and only one same-ladder exception. The rung contains both same-ladder exceptions and at most one opposite-ladder exception. Each therefore has an exception absent from the other.
8. *A rail of one ladder and a rung of the other.* Consider an A -rail in row i and a B -rung with neighbourhood $N_B(q)$. Choose a boundary row different from i and a boundary column outside $N_B(q)$. The corresponding selected turn belongs to the rail but not the rung. Conversely choose $j \in N_B(q)$ so that $T(i, j)$ is selected. If i is a boundary row, every j works. If i is interior, an endpoint neighbourhood supplies a boundary column, while an internal two-column neighbourhood contains exactly one column of the required parity. Then $T(i, j)$ belongs to the rung but is omitted by the rail. The other orientation is symmetric.

The cases include endpoint rungs and both rail sides. A square insertion introduces only new rails and rungs with the same indexed incidences, so it introduces no additional edge type or

boundary case. Thus all distinct edge signatures are incomparable. Apply Proposition 2.4 and equation (10). $\square$

8.3 The exact formula and its lift

Theorem 8.3 (Exact double-ladder formula). *For all odd integers $a, b \geq 3$,*

$$\boxed{\text{hsep}(D(a, b)) = \frac{(a + 2)(b + 2) - 1}{2}}. \quad (12)$$

Proof. The lower bound is Theorem 8.1; the upper bound is Theorem 8.2. Their cardinalities agree. $\square$

Both certificates lift locally under square insertion. For example, under $b \mapsto b + 2$, keep column zero and shift every old positive column by two. The two new columns contribute four boundary cycles and one selected parity cycle in each of the $a - 2$ interior rows, for a net increase of $a + 2$. For the packing, the same strip contributes four boundary singletons and $a - 2$ dominoes. The $a \mapsto a + 2$ rule is symmetric and adds $b + 2$. At the level of graph edges, turns at the expanded cell and two alternating exceptions require the uniform local replacement inside the new square gadget; all other cycles lift by replacing subdivided rails with their three-edge paths.

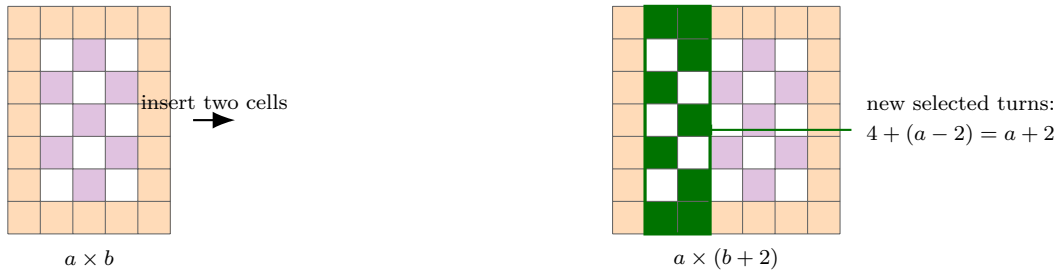


Figure 5: The separating-family lift for $b \mapsto b + 2$, illustrated with $a = 7$. Green cells are precisely the new selected turns: four boundary turns and one parity turn in each interior row. The packing lift has the same increment, with the interior additions represented by domino supports.

The theorem concerns $D(a, b)$ itself. It does not assert that these graphs maximize hsep among all Barnette graphs of their orders.

9 Prospective confirmations beyond order 36

Before the general proofs were completed, a timestamped, read-only prediction record froze the following values before any target graph was constructed or analyzed:

G	$ \mathcal{H}(G) $	$\text{hsep}(G)$
$D(9, 9)$	86	60
$D(11, 9)$	104	71
$D(11, 11)$	126	84.

All six frozen values were subsequently confirmed exactly by explicit primal and matching packing certificates and independent verification; the three graphs appear in Figure 2. Their values now also follow from Theorems 7.1 and 8.3. No complete Barnette census was run at order 40, 44, or 48. In particular, the graph-specific identity $\text{hsep}(D(9, 9)) = 60$ does not imply $M_B(40) = 60$, and no order-level extremal claim is made for any of the three orders.

10 Tied orders and open extremal conjectures

10.1 Tied maximizers in the certified range

The extremal behavior at orders congruent to two modulo four is structurally different from the uniform double-ladder recurrence.

Computational Result 10.1 (Certified tied-order structure). Every one of the 17 tied maximizers at orders 22, 26, 30, 34 contains a 3-cycle component in the face-adjacency graph induced by quadrilateral faces, as well as at least one strict ladder component. The full quadrilateral component signatures are pairwise distinct within each order and reproduce the tie counts 2, 5, 3, 7.

Twenty-five exact square-gadget certificates link these graphs to earlier examples. Thirteen links are literal ladder-cell insertions. In the other twelve, the same four-vertex gadget is inserted on two suitable disjoint edges of a 6-, 8-, or 10-face end-cap region. Thus these graphs are not asserted to be members of one $D(a, b)$ recurrence.

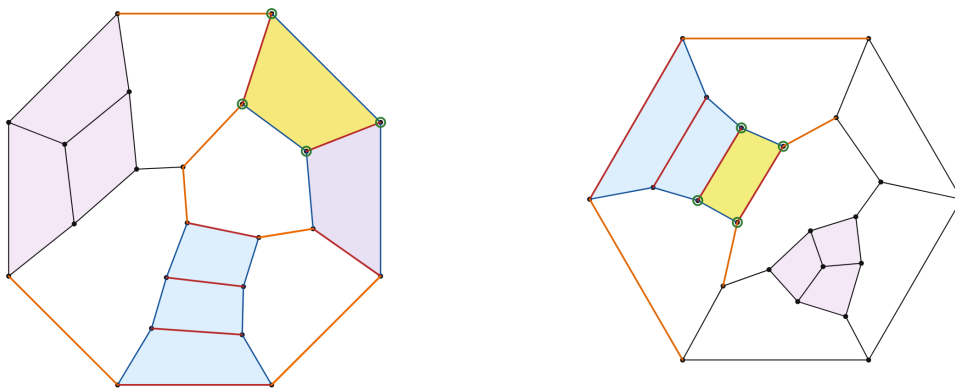


Figure 6: The two certified order-22 maximizers. Purple marks the quadrilateral 3-cycle component; blue and red distinguish ladder rails and rungs. Their different component signatures illustrate why the tied branch is not identified with a single double-ladder skeleton.

10.2 Open extremal continuation

For each order divisible by four through 36, the unique certified maximizer is the balanced or nearly balanced double ladder listed in Computational Result 6.4. The exact value of $\text{hsep}(D(a, b))$ is now known for all odd $a, b \geq 3$, but this does not compare $D(a, b)$ with all other Barnette graphs at the same order.

Conjecture 10.2 (Extremal continuation). *At orders divisible by four beyond the certified census range, the balanced or nearly balanced double ladder is the unique Barnette graph maximizing Hamiltonian edge separation.*

Proving Conjecture 10.2 requires order-level arguments or complete competing-graph certificates; the exact identity (12) alone is insufficient. In particular, this manuscript does not claim $M_B(40) = 60$.

The previously recorded linear fit to the three tied-order values at 26, 30, 34, which predicts $M_B(38) = 39$, remains an especially weak extrapolation rather than a conjecture supported by a transfer proof. No order-38 computation is reported here.

11 Reproducibility and independent verification

Plantri 5.8 generated the complete non-isomorphic Barnette censuses through order 36: 22,263 graphs across the nonempty orders. Retained artifacts include canonical graph records, planar embeddings, cycle universes where required, separating families, and lower certificates.

A standalone Python standard-library verifier reparses the artifacts without project, graph, or optimization libraries and checks graph and cycle structure, ordered edge-pair covers, lower certificates, summaries, and manifests. It does not independently prove Plantri’s enumerative completeness; that premise rests on the recorded complete invocations, contiguous output, and record audits.

The four distinct scopes are certificate verification, the census-completeness premise, software regression, and figure/layout verification. Of 203 current repository tests, 198 passed and five environment-gated Plantri or cubhang integrations were skipped; these skips do not replace certificate checks.

The layouts were initialized from certified planar embeddings using Gunnar Brinkmann’s drawing program and then refined under explicit structural constraints. The abstract graphs, rotation systems, face boundaries, and edge sets were independently verified. Use of the program does not imply endorsement, collaboration, or coauthorship.

The accompanying repository contains the producer, structural analysis, tests, standalone verifier, and manuscript. Large immutable packages remain external; their identifiers, run-times, manifest statistics, prediction lock, layout audit, and order-36 identifier are recorded in `reproducibility_supplement.tex`.

Draft AI-assistance disclosure

AI tools assisted with software development, data analysis, drafting, and language editing. The author is responsible for all definitions, claims, proofs, source verification, and interpretations.

12 Conclusion

The certified computation determines $M_B(n)$ for every nonempty Barnette census through order 36, including all ties and the unique order-36 value 49. Explicit Hamiltonian-cycle families give upper certificates; packings or complete-universe exclusions give lower certificates, all checked by a standalone verifier.

For odd $a, b \geq 3$, the double ladder $D(a, b)$ has one all-rails Hamiltonian cycle, ab turn cycles, and four connector exceptions. Matching symbolic packing and separating-family constructions prove $\text{hsep}(D(a, b)) = ((a + 2)(b + 2) - 1)/2$ throughout this infinite family.

Prediction-locked computations confirm the cycle and separation formulas on three larger family members, but no census-level conclusion follows at orders 40, 44, or 48. Global extremality beyond order 36 remains open, as does a structural explanation of the tied branch at orders congruent to two modulo four.

Acknowledgments

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